

ElasticEdge: An Intelligent Elastic Edge Framework for Live Video Analytics

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Abstract—Cloud computing and edge computing models are popularly applied in emerging applications, such as smart homes, smart parks, and connected autonomous vehicles for large-scale live video analytics. Cloud computing-based models transfer all data to the cloud for video analytics, which burdens network bandwidth and increases the data transmission overhead. Edge computing mode enables video data to be processed at the edge node, thereby reducing the bandwidth overhead. Existing edge computing-based models optimize the performance, but they still have defects in three perspectives: 1) enabling end users to control video content in a real-time format; 2) efficiently locating and transferring the user region of interest (ROI) video data in the video stream; and 3) adapting to various network conditions. To tackle these challenges, we proposed an intelligent elastic edge framework for live video analytics, known as ElasticEdge. ElasticEdge enables the interaction between the end user and the edge node. Elasticity is reflected in two perspectives: 1) the dynamic changes of user requirements and 2) the dynamic changes in network conditions. In addition, ElasticEdge transmits the video stream to the end users based on the tradeoff between the amount of video data and users' ROI to meet various network conditions. To validate ElasticEdge, we conducted experiments to study its performance in comparison to RTFace. The experimental results show that ElasticEdge has a significant edge over RTFace in terms of data transmission. Using 1/16 reserved images, ElasticEdge saves 75% bandwidth and reduces latency by approximately 10% compared with RTFace. We also find that ElasticEdge adapts to various network conditions when streaming videos, i.e., it can reliably obtain essential information with low latency even when the network condition is poor.

Index Terms—Edge computing, filter, live video analytics, region of interest (ROI), video.

I. INTRODUCTION

FUELED by the development of the Internet of Everything (i.e., IoE), video analytics technologies are widely employed to process massive video data [1], [2] in a variety of video services, including the Internet of Vehicles [3], [4], action recognition, smart homes [5], [6], telemedicine [7], [8],

and multimedia IoT [9], [10], live video [11], and game [12]. In a large-scale video stream, cloud computing-based intelligent video analysis of remote video services cannot satisfy the increasing requirements on latency and bandwidth, as indicated by [13] and [14]. The edge computing paradigm [15]–[17] is proposed to address the mentioned problems and improve bandwidth, latency, and storage in cloud-based video data analytics by moving data processing tasks close to data sources. The edge computing paradigm has promising performance.

Compared with the cloud computing model, the existing edge computing-based video analytics systems reduce the bandwidth burden and latency during the video transmission, thereby improving the Quality of Service (QoS). But our work focuses on the challenges and defects in edge computing-based video analytic systems. In an edge video analytics system, video data transmission is a one-way mode (from the camera to the end user); therefore, the end user cannot elastically control the process of video analytics according to real-time demands. This issue is defined as Problem C1 in our work. When the end user can control the video content, it is challenging to automatically select the video data of a region of interest (ROI) in a video stream. This case is referred to as the second problem (C2). The network conditions are essential in video data transmission because the above benefits cannot be achieved when the ElasticEdge does not adapt to poor network conditions. The video service quality cannot satisfy the user requirements due to the poor network bandwidth. We consider this problem as Problem C3 in our work. Enabling end user to control video content in a real-time format requires locating user's ROI. To transfer the valuable data based on the user's preference, we must consider dynamic network conditions to satisfy application requirements.

For example, when natural disasters (e.g., Earthquakes and fires) [18], [19] occur, because of significant damage to the communication infrastructure, it is hard to transfer complete disaster video to be used in the rescue. Therefore, edge video analytics is applied to reduce data volume by adjusting video stream parameters, such as resolution, frame rate, and coding methods. However, existing approaches do not allow the end user to dynamically define ROIs-based user requirements further to reduce video stream volume through a two-way communication model. In addition, video transmission suffers when the network condition is poor. ElasticEdge addresses the above challenges by dynamically obtaining ROI data and adapting to network conditions. ElasticEdge analyzes video data early and transfers critical information transmission to

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the end user in real time. Elasticity is reflected in two perspectives: 1) the dynamic changes of user requirements and 2) the dynamic changes in network conditions.

Compared with cloud computing, ElasticEdge avoids the continuous uploading of high-definition video data to the cloud center, significantly reducing the network overhead and relieving the cloud center's storage and computing burden. Moreover, in comparison with existing video analytics solutions based on edge computing, ElasticEdge enables two-way communication between the end user and edge cameras so that the end user can control the process of video analytics (for Problem C1). It facilitates video data processing and automatically selects user defined ROI in the live video stream, dealing with Problem C2. ElasticEdge supports video data transmission under low network bandwidth to address Problem C3 by identifying and filtering useless information in live video streams. Thus, it transmits valuable information needed by the end user to meet their requirements.

In ElasticEdge, we built an edge computing-enabled video streaming transmission framework. We designed controlled streaming to provide communication services between the end user and the edge device. The edge device can filter non-ROI in a live video stream and perform video analytics tasks. ElasticEdge sends the control parameters of the end-user ROI in the video stream to the edge node. Then, the edge device configures the video stream with the received parameters to process ROI-based video data. The novelty of ElasticEdge is threefold: 1) our proposed ElasticEdge aims to improve the performance of edge video analytics systems based on ROIs in the video stream. Unlike previous research that reduces the communication overhead by adjusting the resolution, frame rate, and coding methods, our approach reduces the communication overhead by examining content in the stream data; 2) we design a novel mechanism to keep the end user in the loop that allows the end user to dynamically define ROI according to the users' requirement and proposed efficient algorithms to detect and capture the ROI-related data at edge devices; and 3) we develop new algorithms that can adapt to harsh network conditions by dynamically obtaining ROI-based video data.

Extensive experimental results show that ElasticEdge can effectively reduce the bandwidth overhead, latency, and power consumption. Furthermore, it can adapt to various network conditions to transmit important information in the video from the edge devices to the end user. Contributions of this study are listed as follows.

- 1) An intelligent framework, ElasticEdge, is proposed for live video processing based on edge computing. The framework interacts with the end user and edge node, exploiting the edge node's intelligent processing on the video data collected by the edge device. The edge node can filter the video based on the received control command and dynamically determine the users' ROI, which achieves more efficient video services with high quality.
- 2) The prototype of ElasticEdge is designed and implemented, consisting of core components, including the image/video acquisition module, video processing module, streaming media service module, and end-user processing module.

- 3) Extensive experiments reveal that ElasticEdge can offer high-quality service for real-time video analytics under harsh networks. Moreover, compared with RTFace [20], the framework exhibits excellent dynamic performance and effectively reduces the dependence on bandwidth and latency in live video streams, thereby ensuring reliable delivery of crucial information in video data.

The remainder of this article is organized as follows. In Section II, we present the motivation of the intelligent elastic edge framework for live video analytics. In Section III, the design of the system is illustrated. The prototype implementation of ElasticEdge is exhibited in Section IV. The experimental results and analysis are displayed in Section VI. The relevant work is presented in Section VII. Then, our study is concluded in Section VIII.

II. MOTIVATION

ElasticEdge is an edge computing-enabled intelligent framework. There is a rich set of scenarios in which our ElasticEdge aims to provide elastic control for real-time video analytics. In this section, we illustrate three use cases that require elastic control for real-time video analytics according to the end-user's ROI under various network conditions. We designed an edge computing-enabled intelligent framework called ElasticEdge, which aims to fill the gap for existing cloud and edge-computing video analytics models have difficulty fulfilling the requirements of these application scenarios.

Case 1 (Telemedicine): Telemedicine exploits the advantages of medical facilities to conduct treatment for patients in remote areas with poor medical conditions. A telemedicine-based system is mostly composed of wearable cameras, smart mobile devices, video streaming servers, and end user of doctors in a hospital. This system is an example of an IoT smart system for medical video analytics. Medical image/video data require a low-latency network; however, in remote areas, the network connection is unreliable compared with that in cities. Thus, it hardly ensures highly reliable video data transmission in telemedicine, i.e., the low network bandwidth causes transmission data to become stuck and blurred. It is necessary to filter redundant information to improve the bandwidth overload under the low-bandwidth network. ElasticEdge aims to guarantee reliable transmission of critical information, crucial to satisfying the requirements of telemedicine applications, especially under harsh network conditions.

Case 2 (Natural Disaster): When Earthquakes, fires, or floods occur, public infrastructure can be significantly destroyed, and local disaster relief cannot provide satisfactory services. Many sensors (e.g., cameras) are equipped on unmanned aerial vehicles (UAVs), referred to as an IoT smart system, to acquire sensor data, such as video data. The computing unit in UAVs can analyze video data. Most video data about a disaster cannot be transferred to the cloud center. People must use a small number of computing resources to analyze video data for object detection and obtain useful information. Thus, video task analysis in edge devices is essential for end users to obtain real-time data at the disaster site. ElasticEdge aims to provide a few resources to collect and

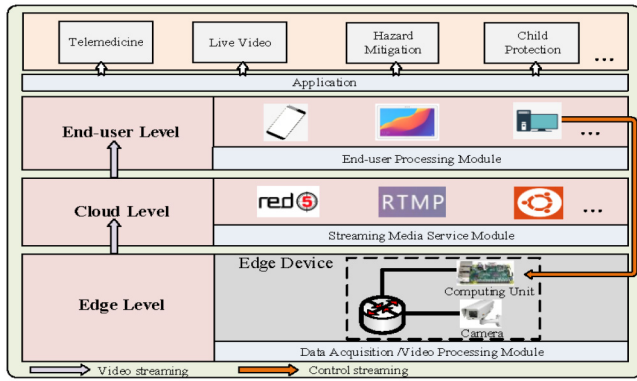


Fig. 1. System overview of ElasticEdge.

process video data at such sites, offering relief workers useful information. Notably, ElasticEdge analyzes video data in the early stages and guarantees critical information transmission to the end user in real time.

Case 3 (Child Protection): With increasing attention being paid to children's safety, many surveillance cameras have been installed for in-school day care. A surveillance camera and nearby a computing unit compose the IoT smart system in school. A surveillance camera-based IoT smart system is applied in a school. A computing unit is embedded in a video camera to run video analytics algorithms in the closed-circuit video surveillance system, such as fall detection and crowd detection. Parents need to know their children's performance and adaptation to the school environment. However, the network overload is enormous when all parents watch videos simultaneously. There is also a privacy issue, because parents can see what other children are doing. In our proposed ElasticEdge, the edge node can be employed to satisfy parents' needs to view their children by filtering out other children. ElasticEdge reduces the overload on the low-bandwidth network on campus. Moreover, ElasticEdge has apparent advantages in child privacy protection.

In summary, in many use cases, computing capability at edge devices and transmission bandwidth on the network cannot meet end-user demands for video processing. It is difficult to achieve reliable transmission of ROI information and real-time video analysis. Thus, we propose an intelligent framework—ElasticEdge—to enable the two-way interaction method between the end user and the edge node with the intelligent processing capability. ElasticEdge transmits the ROI-based video stream by adapting to various network conditions.

III. DESIGN OF ELASTICEDGE

In this section, we present the system overview and design goal of ElasticEdge.

A. System Overview

Fig. 1 presents the system overview of ElasticEdge. ElasticEdge has two data streaming (i.e., video streaming and

control streaming) for live video processing via edge computing. It must be noted that the gray arrow and red arrow represent video streaming and control streaming, respectively.

First, there are four stages in ElasticEdge: 1) data acquisition; 2) video processing; 3) streaming media service; and 4) end-user processing, as shown in Fig. 1. Raw video/image data are acquired in the data acquisition module, which also deals with the low compatibility of equipment, latency, and freeze. The video processing module placed at the edge node conducts video processing and coding to meet the user's demand for video analytics. First, this module elastically processes video data based on the user's ROI in the video stream. Second, the module can dynamically filter video streams when network conditions are poor. The video coding stage determines the user experience and streaming media transmission cost; thus, the coding speed and coding compression ratio should be excellent in video streams. In the streaming media service module, the encoded video stream is pushed to the streaming media server, which delivers the live video stream to the end user. In the end-user processing module, the user defined ROI information is collected.

The edge device is featured with computing capability for video data processing. The edge device comprises an edge computing unit, a video camera (video generator), and a network router. The video camera and the computing unit connect to the router using a wired network. The router sends video data and receives control streaming to/from the upper level. The user equipment used in this study is a fixed device in ElasticEdge. The user equipment connects to the edge device through the wireless network. The edge device connects to the cloud server and the end user through the wireless network. In this study, fixed devices are used as the user equipment. It must be noted that the video camera (or the computing unit) can also connect to the router using the wireless network.

Second, in the video streaming module, the system adopts the Red5 server (Media Streaming Server) [21] as the streaming media server for video transmission streaming. To ensure reliable transmission, the real-time messaging protocol (RTMP) is taken as the transmission protocol. RTMP uses the chunk as the basic unit when transmitting video streams. Developers can customize the size of a chunk. When the first common ACC or H264 [22] is pushed to the server, an audio and video synchronization packet (i.e., audio tag AAC sequence header or video tag AVC sequence header) should be transferred to achieve video synchronization. The common data packet can be sent to the end user via the RTMP protocol when the audio and video synchronization packet is completed. When pushing the H264 stream to the RTMP server, the synchronization package and the H264 data packet guarantee the transmission's reliability. After pushing the H264 stream to the Red5 streaming server via the RTMP protocol, the server produces a uniform resource locator (URL) to publish the video stream. The end user employs the video lan client (VLC) player [23] to play video data by accessing the URL.

Third, we designed the collaborative processing approach between the end user and the edge node in the control streaming module. Thus, the end user can accurately transmit the ROI information to the edge node. The open-source player

VLC is redeveloped to obtain ROI information from the end user. Afterward, the control information streaming from the end user to the edge node should be redefined. The edge node can receive the control information stream sent by the end user in real time and accurately. In ElasticEdge, the video filtering module at the edge device can dynamically filter video data based on the end user's control information (e.g., ROI). In addition, the filter module also elastically processes video streams based on network conditions, thereby reducing the overhead of the data transmission. The two procedures reveal the elastic processing capability of ElasticEdge. Thus, the on-demand information is preserved; meanwhile, redundant video data are eliminated. Finally, the encoded video data are transmitted to the end user.

Summary: ElasticEdge is considered an intelligent framework to address three problems as mentioned in Section I. We employ three modules (i.e., the video processing module at the edge device, the user processing module at the end user, and the control streaming module) to enable the end user to flexibly control the video analysis procedure on demand (Problem 1). The detailed implementation is listed in Section IV-A. We apply the video processing module to relieve the difficulty of automatic ROI selection from videos at the edge level (Problem 2), detailed in Section IV-B. We use the video processing module at the edge level and the video streaming module to address video service quality not meeting user needs under harsh network conditions (Problem 3). Refer to Section IV-C for details.

B. Design Goal

From the three cases presented in Section II, one major goal of edge video analytics is to reduce the amount of transmission while transmitting as much meaningful data as possible. We observe that data transmission is closely related to the ROIs (the interesting data in the video stream). Thus, we propose to study ROIs and their impacts on the amount of video data in data transmission. Therefore, we formalize our research problem based on the volume of transferred data, together with the ROI configuration by the end user and the ROI processing at the edge node. The amount of video data for elastic edge video analytic is defined as follows.

Definition 1: In ElasticEdge, the amount of elastic video data is the size of the user required and the ROI-based portion of a frame in the video stream, which is transmitted to the end user over a period.

With a frame, an ROI is defined by width, height, and origin x and y coordinates. Let α_i represent the area size of the i th ROI in a frame and is determined by the width, the height, and origin x and y coordinates of the ROI. Notation v_i denotes the data size per unit area in the i th ROI. Thus, the data volume of the ROI is represented as $v_i \times \alpha_i$. It is supposed that there is the M number of ROIs in the frame, and we assume that these ROIs are independent in the frame in this work. Data volume of ROIs in the frame can be shown as

$$v = \sum_{i=1}^{i=M} (v_i \times \alpha_i). \quad (1)$$

Let d denote the data volume of a frame; thus, the ratio of the i th ROI data volume to that of a frame is written as $([v_i \times \alpha_i]/d)$. The ratio of ROIs-based data volume to that of a frame is denoted as

$$\beta = \sum_{i=1}^{i=M} \left(\frac{v_i \times \alpha_i}{d} \right), \beta \in [0, 1]. \quad (2)$$

Meanwhile, ROIs-based data volume in a frame can be rewritten as

$$v = \beta \times d, \beta \in [0, 1]. \quad (3)$$

It means that the amount of video data without ROI is $(1 - \beta) \times d$ and the video data cannot be uploaded to the end-user, thereby reducing transmission overhead on the network. Without loss of generality, there is the N number of frames in the time interval $[1, T]$, where the times are expressed in seconds. We use the symbol d_j to represent the data volume in the j th frame. The symbol v_{ij} denotes the data size per unit area in the i th ROI in the j th frame, and α_{ij} is the area size of the i th ROI in the j th frame. The ratio of ROI-based data volume to the j th frame is defined as β_j . Thus, the amount of video data in the range $[1, T]$ is shown as

$$D_{\text{total}} = \sum_{j=1}^{j=N} d_j. \quad (4)$$

Meanwhile, the ratio of all of the ROI-based video data to that of j th frame is listed as

$$\beta_j = \sum_{i=1}^{i=M} \left(\frac{v_{ij} \times \alpha_{ij}}{d_j} \right). \quad (5)$$

In the range $[1, T]$, the amount of ROI-based video data is expressed as

$$V = \sum_{j=1}^{j=N} (d_j \times \beta_j), \beta_j \in [0, 1]_{j \in \{1, 2, \dots, N\}}. \quad (6)$$

Combining (5) and (6), we have

$$V = \sum_{j=1}^{j=N} \left(d_j \times \frac{1}{d_j} \times \sum_{i=1}^{i=M} (v_{ij} \times \alpha_{ij}) \right) = \sum_{j=1}^{j=N} \left(\sum_{i=1}^{i=M} (v_{ij} \times \alpha_{ij}) \right) \quad (7)$$

$$i \in \{1, 2, \dots, M\}, j \in \{1, 2, \dots, N\}.$$

We find that the amount of video data is determined by the area size of ROIs (α_{ij}) in the candidate frame in the video stream. In the ElasticEdge, we can elastically adjust transmitted data volume based on the size of ROI and obtain the ROI-based video data for elastic edge video analytics. Thus, ElasticEdge reduces bandwidth overhead and improve latency under harsh network condition.

The area size of ROI is dependent on the user's requirement for the video stream. End users can preset many ROIs independently according to their width, height, and origin x and y coordinates in a frame. A ROI area is referred to as a ROI feature in this work. In the ElasticEdge, the end-user's flexible control module and a control execution module on the edge node interact with the two modules. The end user transmits control information to the edge node through the flexible

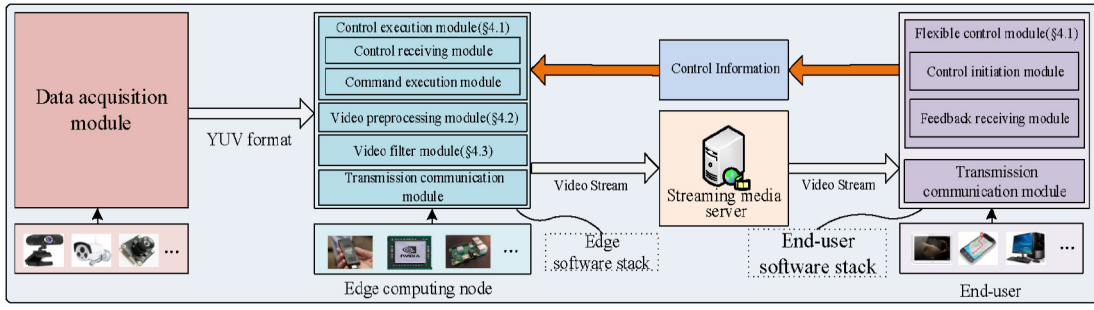


Fig. 2. Implementation of ElasticEdge.

control module. The control information includes: 1) a switch to control the edge node to elastically detect the user's ROI in a frame and 2) the feature vector set of ROIs.

Let symbol l_i represent the feature for an ROI in a frame, and we have $l_i = (w_i, h_i, x_i, y_i)$ in which w_i and h_i denote width and height of the ROI. We employ a feature vector that includes many ROI features. We employ symbols $l_1, l_2, \dots, l_n$ to denote the features of the n number of ROIs in a frame. Then, we have $L = [l_1, l_2, \dots, l_n]$ in which L is the feature vectors set. The format of control flow can be shown as $[c; l_1, l_2, \dots, l_n]$. The metric c is the switch flag. When the switch is triggered to detect ROI, there is $c=1$; otherwise, we have $c = 0$. The symbol c can protect private data. The flexible control module sends this information to the control execution module at the edge node. When receiving the control flow, the execution module detects ROIs. The edge node can elastically select ROIs within frames in the video stream based on the feature vector set. After the end-user requirements are satisfied, a flexible control module sends a signal to turn off elastic detection to the execution module at the edge node.

At the edge node, the feature vector of ROIs is obtained in advance according to the end-user's requirement. The edge node can elastically select the user-defined ROIs in the video stream according to the feature vector set. It is assumed that the end user turns on object detection. The edge node can obtain the m number of frames from the camera. We have

$$F = [f_1, f_2, \dots, f_m] \quad (8)$$

in which f_i denotes the i th frame in the video stream and F means the tested frame set. The end user initially sends a control flow to the edge node, turning on the elastic detection, and sequentially processing the frames based on the feature vector set. Then, we have

$$\begin{aligned} F^T \cdot L &= \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_m \end{bmatrix}_{m \times 1} \cdot [l_1 l_2 \dots l_n]_{1 \times n} \\ &= \begin{bmatrix} f_1 \cdot l_1 & f_1 \cdot l_2 & \dots & f_1 \cdot l_n \\ f_2 \cdot l_1 & f_2 \cdot l_2 & \dots & f_2 \cdot l_n \\ \vdots & \vdots & \ddots & \vdots \\ f_m \cdot l_1 & f_m \cdot l_2 & \dots & f_m \cdot l_n \end{bmatrix}_{m \times n} = \begin{bmatrix} f'_1 \\ f'_2 \\ \vdots \\ f'_m \end{bmatrix}_{m \times 1} \cdot \quad (9) \end{aligned}$$

The edge node sequentially acquires frames and outputs a set of ROIs-based images in real time. For example, the frame f_i is

processed by feature vector l_i ; then, we have $f'_1 = (f_1 \cdot l_1, \dots, f_1 \cdot l_i, \dots, f_1 \cdot l_n)$ in the frame. Then, we achieve ROI-based video data composed with the processed frames, reducing bandwidth overhead and latency between the end user and edge nodes. It must be noted that the ROI feature l_i in the frame determines the minimum area α_i in the candidate ROI. Besides, there are no ROIs in the frame when the feature vector set is empty.

IV. ELASTICEDGE: PROTOTYPING

The prototype of ElasticEdge is built to achieve real-time video processing. In Section IV-A, the end-user component is redesigned to enable users to control video by using two-way communication, based on which, the interaction between the edge node and the end user is designed instead of passively accepting video data for the end user. In Section IV-B, the video analysis task is preprocessed at the edge node to select ROI video data automatically. A video filtering method is implemented (detailed in Section IV-C) to improve video services in harsh environments.

A. End-User Interaction With the Edge Device

This section aims to solve the end-user's problem of lacking the ability to control the delivered video content in real time. The main challenge lies in how to support end user and edge node interaction. Next, we describe how we enable end users to set up ROI in video data in a real-time format.

We redeveloped VLC to capture user ROI information. Through libvlc in VLC, the source code is modified to customize the display interface and capture users' ROI in the video stream. For example, in intelligent transportation, the end user can mark a predefined lane for the departure warning at an intersection as ROI information. Then, the customized VLC is designed to collect the ROI information from end users. The VLC can extract the video region location containing the predefined lane; then, VLC sends the ROI information to the edge node in a control stream. TCP-based data packets are used to ensure reliable transmission between the end user and the edge node.

As shown in Fig. 2, the control module in ElasticEdge is composed of a flexible control module at the end user and a control execution module at edge nodes. The two modules conduct collaborative video data analytics in ElasticEdge. They achieve the interaction between the end user and the edge node. The control information is sent to the edge node by

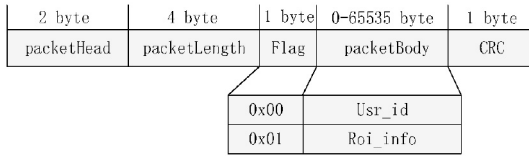


Fig. 3. Customize the packet format.

the end user. In the flexible control module, the control initiation module and feedback receiving module are responsible for starting control information and sending control information to the edge node. In the control execution module, the control receiving and command execution modules are responsible for receiving the control information from the end user and executing the control information. We analyze the implementation of control functions in the following two scenarios.

Case 1: The end-user sends commands to the execution module at the edge node.

Case 2: The edge node automatically controls the video data processing without the end-user commands.

The control stream primarily targets transmitting the end-user's ROI in the video stream to the edge node processing module. The communication between the end user and the edge node is built based on the TCP protocol. As shown in Fig. 3, we defined the packet data format as (packet header, length, and body). The server can combine packets as a transport protocol to reorganize the receiver data. Furthermore, we design a transmission control protocol for ElasticEdge. A connection-oriented client-server model is established. The end user is set as the client, while the edge node is referred to as the server. The location information of the ROI is configured as the event stream. The edge device waits for requests from the end user. The end user can reliably deliver the information of user-defined ROI to the edge device promptly.

B. Video Preprocessing

The challenge is how to preprocess the video data at the edge device to select ROI from the video stream automatically. Fig. 2 shows that the video data are processed in the video preprocessing module at the edge node. The YUV-based raw video data cannot be encoded and transmitted; thus, video preprocessing, including video enhancement, image splicing, and video superposition, is carried out. The raw video data in YUV [24] are converted to RGB format through a per-pixel transformation formula to facilitate the video data adjustment. Image denoising, image reproduction, image segmentation, and feature extraction are used for preprocessing RGB video data, and ElasticEdge can select video data according to user-defined ROI in the video stream.

C. Edge Computing-Based Video Filter

The video filter in ElasticEdge can handle the problem that the video service quality cannot satisfy the application requirements under the low-bandwidth network. The challenge in this work is to filter the unnecessary data in the video stream efficiently. The video-filtering component filters the video by an edge computing node, reducing redundant information in the

video data. When accurate ROI information is sent to the edge node by the end user, the data processing module stops transmitting the whole video data to reduce the bandwidth cost. The video filtering process is conducted before video coding. Fig. 2 shows that the video filtering module at the edge node conducts the video filtering operation. The video filtering module delimits the ROI in the video stream that comes from the video preprocessing module. Based on the user-defined ROI, the ROI in the video data is extracted to be encoded and transmitted to the end user, and the non-ROI and redundant video data are filtered, which can reduce the amount of transmitted video data.

D. Libraries for ElasticEdge

When implementing the prototype of ElasticEdge, we employed standard libraries for video encoding, streaming media protocol, streaming media server, and multimedia player.

Video Encoding Method: There is heavy data redundancy in the video stream. Thus, ACC or H264 [22] is adopted to synchronize the audio and video, removing redundant video data. H.264 is the most coding standard for a video stream. H.264 can achieve higher coding efficiency and high-quality video stream at a low bit rate than H.263; moreover, it supports the real-time video data transmission because of its lower complexity, compared with H.265. We reconfigure the preset values of length, width, frame rate, and bits in the video stream at the edge node in our work. The goal is to tradeoff encoding speed and video quality, including: 1) obtaining high-quality video data with limited bandwidth and 2) reducing the video data volume without sacrificing video quality.

Streaming Media Protocol: RTMP [25] maintains a persistent TCP connection between the video player and the Red5 streaming server, delivering a reliable and lightweight stream to the end user. RTMP uses the chunk as the basic unit when transmitting video streams. Edge nodes send video streams in chunks sequentially. The server assembles the original stream from edge nodes based on chunk ID and header. The RTMP protocol can retransmit the data packets that cannot be successfully transmitted to end users, avoiding the loss of data packets. The RTMP protocol provides real-time video stream transmission, reducing network bandwidth limitation on video transmission. Thus, we employed RTMP in ElasticEdge.

Streaming Media Server: The streaming media server is regarded as an information transfer station, which is also used to play video streaming. As an essential platform for video services in a natural disaster, the latency and quality of the streaming media server are particularly important. Our ElasticEdge employs RED5 [21] to parse and verify the H.264 streams of the RTMP protocol and then produce URLs and publish them to the streaming media player client at the end user. The end user can use the URL to watch the live video stream.

Multimedia Player: In ElasticEdge, we employed a popular video player—VLC [23] as a flexible control module in ElasticEdge. We select VLC because VLC is an open-source, free, cross-platform media player. We added the user-defined

ROIs module and ROI transmission module based on the secondary development on VLC. End users utilize the back-end control module developed based on VLC to customize the ROI.

E. Summary

In ElasticEdge, the critical information in the video stream is filtered out and transferred to the end user. The advantages of ElasticEdge include twofolds. First, the source video is processed according to the interest of the end user. The video filtering module can obtain the ROI and send it to the end user. Second, the amount of transmitted data volume decreases, thereby improving the video transmission latency. In real-time scenarios, the approach to filtering video data can effectively improve the real-time performance during data transmission. We validate the advantages of our proposed ElasticEdge in the following sections.

V. DISCUSSION

In this section, we discuss the design of ElasticEdge in this work, including design tradeoffs, mathematical model, extended application of ElasticEdge, and future work.

A. Design Tradeoff

First, in the ROI of the video stream, different feature vectors for the detected objects are needed to meet the requirements of accuracy in video analytics. High-accuracy video analytic tasks use high-dimensionality feature vectors, containing more video data, but network bandwidth and computing resources on the edge device are consumed. We should trade-off the size of the feature vector and the cost of the resources. Second, there are different types of ROIs based on the requirements of applications (e.g., face detection, action detection, and passenger counting), but more ROIs-based video processing costs more resources on the edge device and increases the latency for end users. There is a tradeoff between the ROIs (no-ROI versus ROIs, ROIs, and critical ROIs) in the video stream and resource cost on the edge device. For example, some ROIs should be discarded under a harsh network compared with the critical ones. An end-user faces the tradeoffs in the number of ROIs, ROI types, accuracy, computing resources cost, latency, and network bandwidth.

B. Mathematical Model

In this work, ElasticEdge aims to transmit the required video stream (or user-defined useful video stream) to end users based on the tradeoff between the amount of video data and the ROIs of users to meet various network conditions. We defined the bandwidth model of ElasticEdge based on users' ROI in the video stream. Refer to Section III-B for details on the model. This model reveals that ElasticEdge can improve the bandwidth cost by using fewer resources and reducing data transmission overhead under harsh network conditions.

C. Extended Application of ElasticEdge

Our proposed ElasticEdge can be extended to other IoT systems assisted by intelligent edge, such as a smart industrial

park, forest fire rescue, and emergency medical services. We can design customized smart sensing devices, e.g., a video camera equipped with a computing unit. Then, the ElasticEdge can be deployed on smart sensing devices to enable end users to obtain the required video data in real time with low network bandwidth and limited computing resources.

ElasticEdge offers real-time ROI-based video data to decision-making module under the mobile network. Thus, it can be applied in connected and automated vehicles (CAVs) networks, where CAVs interact with other vehicles and environments to collect a large amount of data in real time and make driving decisions, thereby reducing the potential collision and improving traffic safety.

In the glory of the king (a popular game), Ma Chao (player P1) needs to track the competitor (Guan Yu)—player P2—in an area. Player P1 uses ElasticEdge to define the area as ROI and detects whether player P2 leaves the area. Player P1 is alerted if the player P2 leaves the preset area. The video data are from video files rather than external video cameras. Different regions of interest in the game can be defined according to the user's requirements.

Augmented and virtual reality (AR/VR) glasses for inspection and maintenance processes are widely used in smart factories. A glass worn by maintenance personnel can be deployed with our ElasticEdge. Maintenance personnel can install or repair the key components in the equipment. The technical experts can supervise this process in the background. To check if key components are correctly installed by maintenance personnel, the expert can set the regions of key components in the equipment as ROI using ElasticEdge on the AR/VR glass. If a component is installed incorrectly, the expert uses ElasticEdge to send the installation data and alarm information to the AR/VR glass, assisting the maintenance personnel in work.

D. Open Research Issues

In this section, we discuss open research issues in ElasticEdge, including privacy protection, adaptive ROIs definition, and collaborative edge cache management.

First, in the current ElasticEdge design, the transferred video stream in the format of ROI may contain user private information [26]. Novel privacy protection methods are essential to protect the privacy of video data in ElasticEdge. For example, the surveillance video stream from the camera can contain confidential information (e.g., personal and sensitive data) that may not be revealed to the end users. If the private data remain in targeted ROI, on-site privacy-preserving approaches for video analysis should be designed in ElasticEdge, such as deep learning-based video filtering to remove privacy information (human face) [27], [28], denaturing video streams that selectively blurs sensitive regions according to end-user requirements [29], and multilevel privacy protection schemes in video streams to match edge devices hardware and video contents [30].

Second, the adaptive ROIs definition is determined by dynamically changing environments and different end-users requirements [31]. To cope with the dynamic environment,

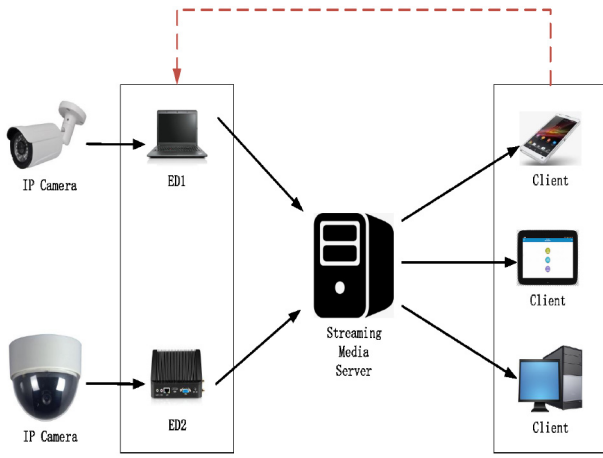


Fig. 4. Testbed of ElasticEdge.

adaptive ROI definition methods should be designed to capture the ROIs in the video stream according to different scenarios and end-user requirements. For example, when the network bandwidth is poor, deep learning-based ROI definition methods [32] can be used to dynamically capture the ROI size, which adapts to the harsh network environment. End users may be interested in different ROIs; therefore, we can design on-demand content-based approaches in ElasticEdge to achieve the adaptive ROIs [33] to meet the requirement of each end user.

Third, video stream typically demonstrates a significant spatiotemporal correlation in a cameras-dense video deployment system [34]. A ROI can be composed of multiple sub-ROIs from more than one camera. The integrated multiuser-defined ROIs in multistreams from different cameras can avoid isolated camera deployment and improve detection accuracy [35]. We can design a collaborative edge cache management [36] to conduct the integration ROIs in ElasticEdge. For example, a collaborative edge cache management framework can be deployed in an intelligent gateway, which connects to multiple collective cameras through networks. The edge computing-empowered intelligent gateway can integrate multiple ROIs offered by more than one camera into an ROI. Using the collaborative edge cache scheme, we can design deep learning models-based collaborative edge cache schemes [37], multisource data fusion methods [38], and cross-camera collaboration schemes [39] to conduct real-time video analytics in ElasticEdge. End users can achieve the high cache hit ratio and low access latency for the object detection, thereby improving the QoS.

VI. PERFORMANCE EVALUATION

We conducted experiments to validate our proposed ElasticEdge on a real world-tested platform.

A. Experimental Setup

Tested Platform: Fig. 4 presents the deployment environment of ElasticEdge to validate its performance in terms of bandwidth, latency, and energy consumption under different networks. The testbed comprises cameras, an edge computing

unit, a video stream server, and clients (end users). The camera and an edge computing unit are referred to as the edge device. Then, edge-side ElasticEdge is deployed on the computing unit to preprocess video streams in real time. The end user of the ElasticEdge is deployed on a server. The live streaming media server RED5 is installed on a server named the video stream server in our platform. The video stream server receives the video data and sends it to the client (end user) in real time. The client supports defined ROIs and transfers ROIs between the client and the edge device. We used two edge computing units, including a laptop and an industrial computer denoted as symbols ED1 and ED2, respectively. ED1 is equipped with a Core i5-7300 CPU @2.50 GHz and 8-GB memory. ED2 is configured with Celeron J1900 CPU @ 2 GHz and 4-GB memory. ED1 represents an edge node with high computing power, while ED2 simulates an edge node with low computing power.

Tested Performance Metrics: We use three types of video resolution (i.e., 1920×1080 , 1280×960 , and 640×480) and three frame rates in video steaming (i.e., 15, 20, and 25 fps). We employed three-type network connections, Wi-Fi, 4G LTE, and 5G, in the experiments. We select the scene with small fluctuation in the tested video stream to validate the filtering module. We evaluate the performance of the proposed framework in terms of the three metrics defined as follows.

Transmission Bandwidth: Transmission bandwidth refers to the network bandwidth for video data transmission in the video stream. This metric reflects the amount of data transmitted per unit of time.

Latency: The term *latency* represents the time interval between capturing the video for an analytic task at an edge camera and returning the feedback from video analytics tasks to the end user. This metric is dependent on the procedure of video processing and transmission in the video stream. The low latency indicates the high performance of the video analytic system.

Power Consumption: Power consumption quantifies the energy consumption per unit of time on an edge device when performing video analytics tasks.

By altering the resolution and frame rate, we can obtain different video qualities. We also study the impacts of the size of the filtering area in a frame and the type of network connections on the bandwidth, latency, and power consumption.

B. ElasticEdge Versus RTFace

In this section, we compare the performance of ElasticEdge with RTFace [20] in terms of transmission bandwidth and latency. RTFace is another famous live video analytics framework that aims to process and transmit real-time videos based on edge computing.

1) *Transmission Bandwidth:* The ElasticEdge significantly saves transmission bandwidth costs in the case of various video resolutions, sizes of the reserved region, and frame rates in the video stream.

This test aims to validate the performance of ElasticEdge in terms of transmission bandwidth under network conditions with no limited bandwidth. Fig. 5(a)–(c) presents

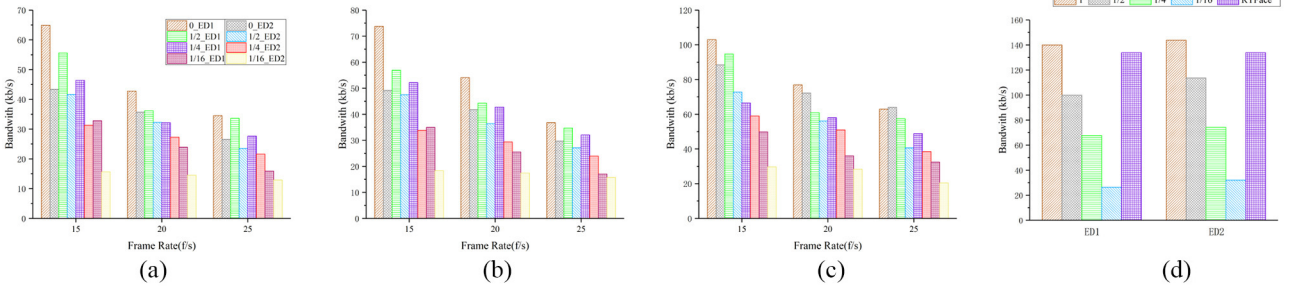


Fig. 5. (a), (b), and (c) Show the key factors that impact the bandwidth based on ED1 and ED2. The x- and y-axes represent the frame rate and the bandwidth, respectively. (d) Shows the comparison between the ElasticEdge and RTFace in terms of bandwidth on two edge devices.

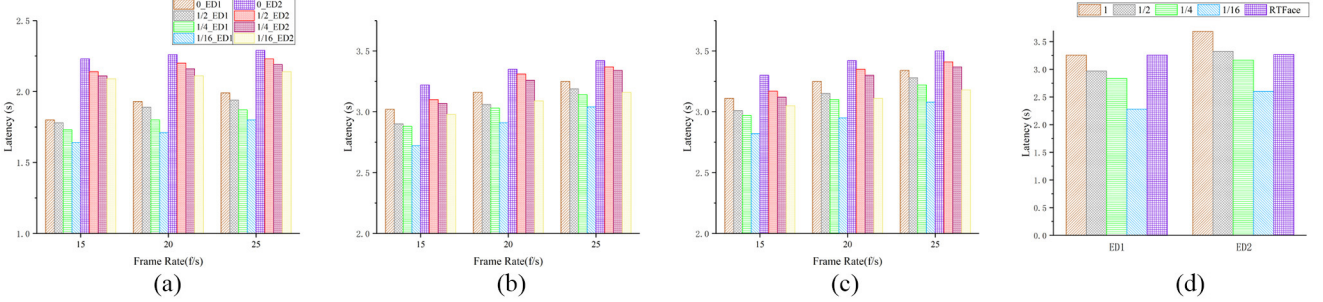


Fig. 6. (a), (b), and (c) Show the influence of different parameter settings on the latency based on ED1 and ED2. The x- and y-axes represent the frame rate and latency, respectively. (d) Shows the latency of ElasticEdge and RTFace on two edge devices.

the transmission bandwidth under three resolutions on edge devices ED1 and ED2. With the same resolution, we set three frame rates (15, 20, and 25 fps) and four reserved region ratios (no filtering, 1/2, 1/4, and 1/16). The reserved region ratio is the percentage of ROI in the overall video frame. The ROI is predetermined to validate the accuracy. The results show that the small reserved region reduces the amount of transmitted data; thus, ElasticEdge significantly decreases the transmission bandwidth costs in the live video stream. With the same reserved region and frame rate, the high resolution causes a huge amount of transmitted data, thereby aggregating the bandwidth overhead. In addition, with the same resolution and filtering region, the high-frame-ratio video stream brings a high computation cost for video data processing but transmission data reduces, leading to decreased transmission bandwidth. In the best case, ElasticEdge with 1/16 of the reserved region reduces the transmission bandwidth by approximately 60% compared to that not using ElasticEdge on two edge devices.

As shown in Fig. 5(d), we compared ElasticEdge with RTFace [20] in terms of transmission bandwidth under the same resolution and frame rate. ElasticEdge has a similar bandwidth as RTFace when the no-filtering module is applied in ElasticEdge. When the reserved region ratio of ElasticEdge is set as 1/2, 1/4, and 1/16, the transmission bandwidth of ElasticEdge decreases accordingly, and the value is smaller than that of RTFace. For example, ElasticEdge takes up approximately 25% of the transmission bandwidth compared with RTFace when the reserved region ratio is 1/16 on two edge devices.

2) Latency: In this section, we study the latency of live video processing in ElasticEdge on edge device ED1 and ED2 in the case of different frame rates, resolutions, and reserved regions. As shown in Fig. 6(a)–(c), the small reserved region after video filtering results in short latency while high resolution causes long latency. The video filtering module in ElasticEdge reduces the amount of transmitted data, thereby improving the latency. For the same reserved region and frame rate, high resolution leads to high latency because much video data needs to be processed, which enlarges latency under a high resolution. Moreover, we can find that the frame rate primarily impacts latency. With the same resolution and filtering region in the video stream, a great frame rate can cause a high latency because it cost much time to process video data. Under low resolution, the latency is 1–2 s, which is caused by video data processing and the RTMP protocol used in ElasticEdge. The RTMP protocol ensures transmission stability at the expense of performance. ElasticEdge has better latency advantages than that without ElasticEdge, significantly reducing latency by approximately 10% when 1/16 of the region is reserved on edge device ED1 and ED2. Unlike ED1, the latency of ED2 is slightly longer when the resolution, frame rate, and filtering area have the same configuration. The reason is that the performance of ED2 is low, leading to a longer execution time. We also compared the latency of ElasticEdge and RTFace with the same resolution, frame rate, and video parameters. As shown in Fig. 6(d), ElasticEdge without filtering has a similar latency as RTFace. When the reservation region ratio is set as 1/16 in the filtering module, both ED1- and ED2-based ElasticEdge have an edge over

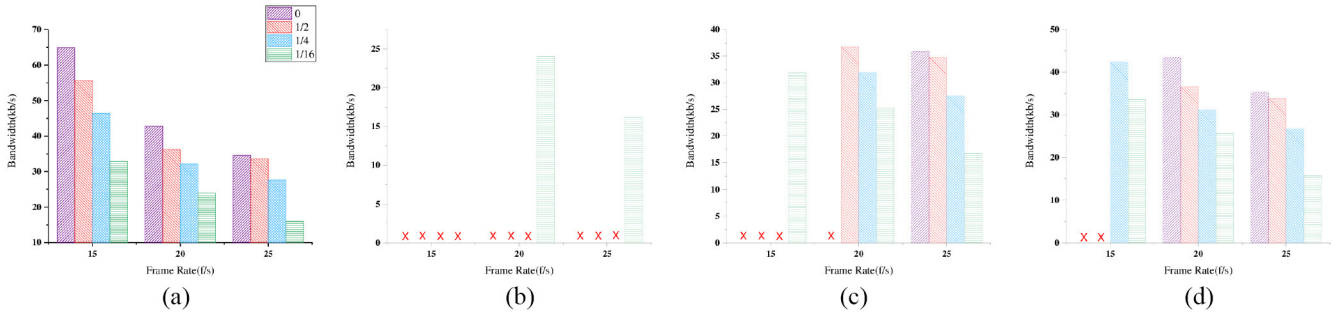


Fig. 7. (a)–(d) Show the influence of different configuration parameters of ElasticEdge on the transmission state under the constraint of network bandwidth. The resolution is 640×480 . The x- and y-axes represent the frame rate and bandwidth.

RTFace in terms of latency. ED2-based ElasticEdge has more advantages than that of ED1 environment.

C. Extended Study

In extended study, we study the performance of ElasticEdge in harsh network conditions. RTFace is not compared here because ElasticEdge has apparent advantages over RTFace with a filtering module for redundant information in video streams.

1) *Harsh Network Condition*: Advances in communication technologies, such as 5G wireless and Wi-Fi 6, improve transmission latency and throughput of the high-definition video stream, but harsh network conditions will remain even though these technologies are used. For example, the harsh network condition will exist in the scenarios of fast-moving CAVs, natural disasters, live broadcast of games, and large-scale IoT devices. Below are examples that illustrate how harsh network conditions occur.

Fast-Moving CAVs: The wireless connection becomes unstable and weak because of high mobility [40], [41]. When the CAVs move fast, especially in a city surrounded by many skyscrapers or in a remote area with low coverage of cellular tower, the bandwidth of CAV-infrastructure communication will be extremely low [42], [43].

Nature Disaster: Natural disasters (e.g., Earthquakes and fires) cause significant damage to the communication infrastructure. Mobile sites may be set up to provide alternative networking services. They usually can only provide low bandwidth, which will be shared by several heavy applications, such as video analytics.

Live Broadcast of Games: Steamers broadcast the live game to their subscribers through 4G/5G cellular networks [44]. They have to share bandwidth with thousands of cellular users in the arena who are stimulatingly using 4G/5G connections. Therefore, the average available bandwidth for each user becomes minimized.

Large-Scale IoT Devices: IoT devices usually communicate through low-bandwidth communication channels, such as Bluetooth [45] and Zigbee [46]. Moreover, when the low-bandwidth wireless channel is shared by multiple parties in a complex environment, the valid bandwidth goes even lower.

One prior work [47] adopted a similar experiment setting like us to simulate low-bandwidth network conditions and then investigate algorithms for selecting optimal system parameters.

Network bandwidth limitations in harsh environments require efficient approaches to meet QoS, such as latency and reliability. Edge computing is one of latest technologies to achieve the above goals; however, there are still some issues existing in the edge video analytics, as mentioned in Section I. Thus, we proposed ElasticEdge to address unsolved challenges, and we set experiments using low bandwidth to simulate harsh network conditions when validating the efficiency of ElasticEdge. We study the transmission bandwidth cost of ElasticEdge using four-type network bandwidths: 1) no-limit; 2) 200 kb/s; 3) 300 kb/s; and 4) 400 kb/s, the last three of which simulate the bandwidth of harsh network conditions. Fig. 7 presents the results of our extended experiments on ED1 under the resolution 640×480 .

In Fig. 7(a)–(d), when the bandwidth exceeds the limited one, without ElasticEdge, the video analytic systems cannot send video streams to end users under the low network bandwidth; thus, the value of bandwidth is zero, which is represented by symbol “ $\times$.” It means that the transmission is unavailable in the video stream. It must be noted that Fig. 7(a) has the same tested values as those in Fig. 5(a). We employ Fig. 7(a) because the result is used as a baseline to compare with the limited network conditions. When applying ElasticEdge to filter the video stream, the video stream can be smoothly illustrated under a low network with the 1/16 region reserved [Fig. 7(b)]. Essential information can be obtained from the video stream. The results suggest that our proposed ElasticEdge can provide advantageous performance under a low-bandwidth network.

2) *Comparison Between Networks 4G and 5G*: In this section, we performed tests to validate that ElasticEdge can easily adapt to different network conditions. We set the resolution to 1920×1080 and three frame rates (15, 20, and 25 fps). There are four reserved regions, no filtering, 1/2, 1/4, and 1/16, to study the bandwidth cost under 4G and 5G networks, respectively. In the 4G network, we used a Qualcomm Snapdragon X24 LTE modem. The download speed (peak) is 2 Gb/s. In the 5G network, we employ HiSilicon Balong 5000 modem. The download speed (peak) is 4.6 Gb/s. The details of 4G and 5G characteristics are listed in Table I. We use a network monitor—net-peeker that installs on the edge device to record bandwidth under networks 4G and 5G.¹ The results are shown

¹Net Peeker. <http://www.net-peeker.com>.

TABLE I
CHARACTERISTICS OF NETWORKS 4G AND 5G IN THE TEST

Mobile Networks	4 G	5 G
Company	Qualcomm	HiSilicon
Modem chipset	Snapdragon X24 LTE	Balong 5000
Download speed (peak)	2 Gbps	4.6 Gbps
Upload speed (peak)	316 Mbps	2.5 Gbps

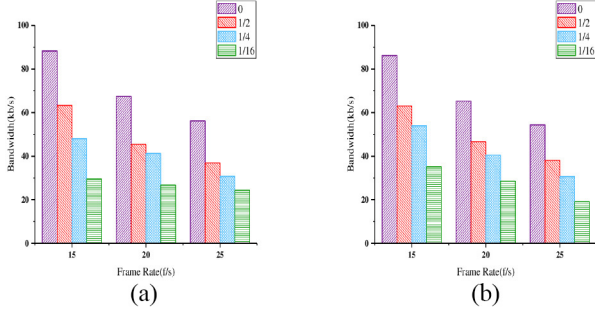


Fig. 8. (a) and (b) Show that ElasticEdge improves bandwidth cost under networks 4G and 5G.

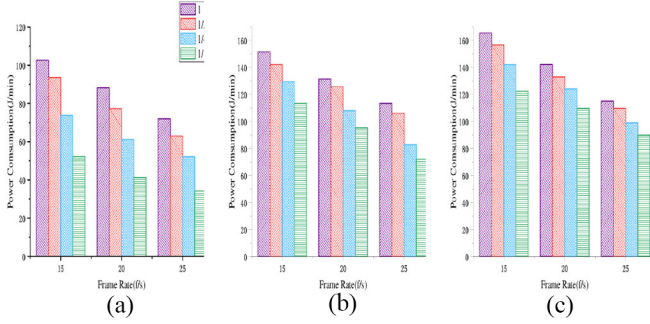


Fig. 9. (a)–(c) Show the main influencing factors of power consumption under ElasticEdge, and the x - and y -axes represent the frame rate and power consumption.

in Fig. 8 In Fig. 8(a) and (b), comparing with wired networks, it can be found that the small filtering reserved region costs a low bandwidth under networks 4G and 5G. The reason is that the small transmission region reduces the data volume, and then the bandwidth utilization becomes low. Thus, ElasticEdge can significantly reduce the bandwidth cost under different networks.

3) *Power Consumption*: Devices at the edge are extremely sensitive to power consumption. The power consumption of ElasticEdge is tested with different frame rates, resolutions, and reserved regions in the video stream. A power recorder designed by ZHURUI-TEC² measures the energy consumption of the edge device. We record the execution time in the edge; and then, we can obtain the power consumption, as shown in Fig. 9. With the same resolution and frame rate, the small reserved region can lower the power consumption because video stream filtering reduces video data and power consumption. When the resolution increases, the power consumption increases with the same frame rate and reserved region. Under

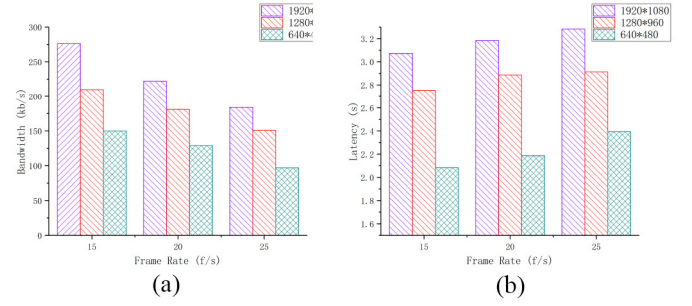


Fig. 10. (a) Bandwidth and (b) latency of ElasticEdge in the multi-ROI scenario.

the same resolution and filtering region, the high frame rate costs great power consumption because high resolution and large frame rate video streams consume many computing resources. The application of ElasticEdge can significantly reduce power consumption.

4) *Multiple ROIs-Based Scenarios*: There may be many ROIs required for end users in the video stream in realistic scenarios. In this test, we modified ElasticEdge to support multiple ROIs. We employ the edge device ED2 in the experimental setup. Without loss of generality, we exemplify the multiple ROIs as three reserved regions in a frame, one of which has the same reserved region ratio. Fig. 10 presents the performance of ElasticEdge in terms of bandwidth and latency under multi-ROIs scenarios. Compared with the single ROI with 1/4 and 1/2 reserved regions, the bandwidth of ElasticEdge with multi-ROI (3/8 reserved region) decreases by 6.7% and 24.4% [see Fig. 10(a)], respectively. On edge devices, many ROIs consume more computing resources to process video data locally, which increases the latency of video processing [see Fig. 10(b)]. However, the latency of ElasticEdge with three ROIs increases slightly. Compared to ElasticEdge with a 1/2 reserved-region ROI, ElasticEdge with multiple ROIs increases 2.67% latency under resolution 1920×1080 and 25-fps frame rate. In other test cases, the latency of ElasticEdge presents smaller than that of the scenarios with 25-fps and resolution 1920×1080 . The experimental results reveal that it is worth sacrificing less latency to achieve more ROIs in the video stream.

5) *Multiple Edge Devices and Multiple User-Based Scenarios*: We study the performance of ElasticEdge with multiple edge devices and multiple end users (see Figs. 11 and 12). This test employs three ED2s, three video solutions, three frame rates, four types of reserved ROIs, and three end users. Video data from the three edge devices are transmitted to the video stream server. Each end user can obtain video data through URLs in RED5 on the video stream server. We compare the bandwidth and latency of the multiple edge devices and end users-based ElasticEdge (MEE-ElasticEdge) with a single edge device and end user-based ElasticEdge (SEE-ElasticEdge). In Fig. 11, the video resolution has little impact on the bandwidth. For example, with 15 fps and no-ROIs, the bandwidth of MEE-ElasticEdge decreases by 15.78%, 14.07%, and 12.87% under resolutions 1920×1080 , 1280×960 , and 640×480 , respectively. Although more

²<https://shop60612912.taobao.com/index.htm?spm=2013.1.w5002-23043798976.2.6318680f9134a6>

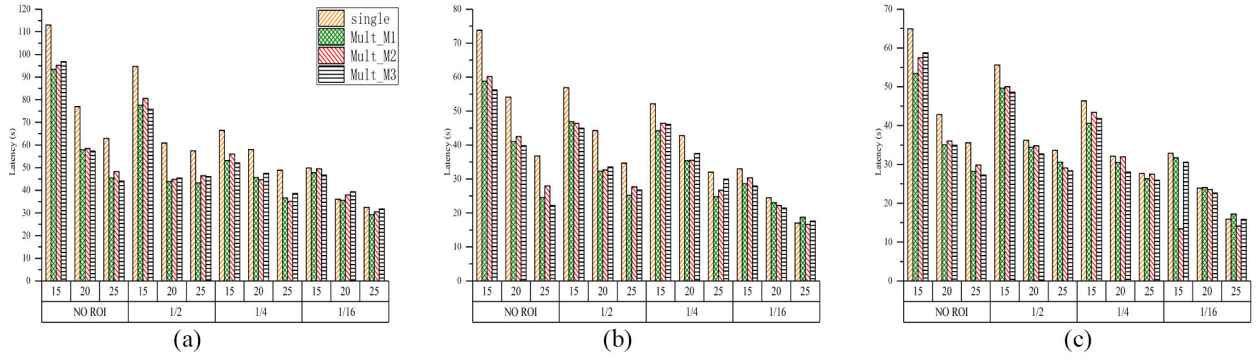


Fig. 11. Bandwidth of ElasticEdge with multiple edge devices and multiple end users. (a) 1920×1080 . (b) 1280×960 . (c) 640×480 .

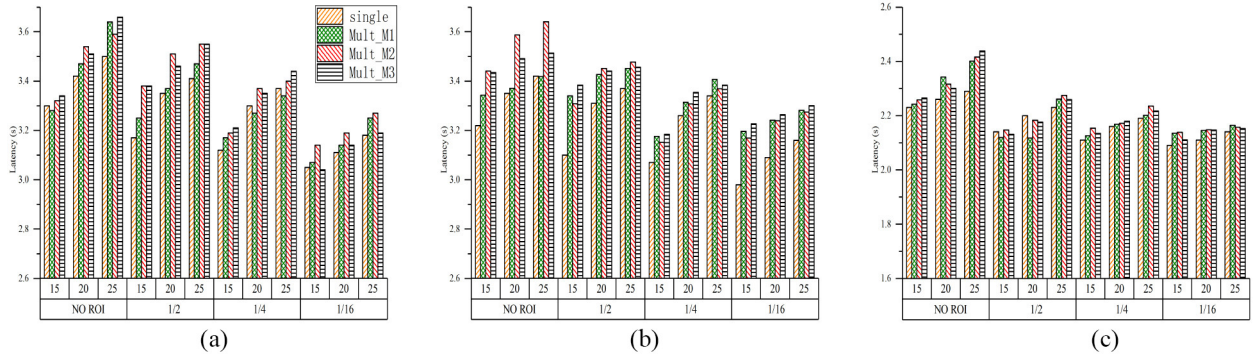


Fig. 12. Latency of ElasticEdge with multiple edge devices and multiple end users. (a) 1920×1080 . (b) 1280×960 . (c) 640×480 .

edge devices aggregate the load of RED5 compared with a single edge device, it has few side effects to bandwidth and latency. The bandwidth slightly drops with a low resolution and small-size ROI. Without ROI, the average latency of MEE-ElasticEdge increases by 3.6% at maximum under resolution 1920×1080 and 25-fps frame rate. With the reserved region ratios 1/2, 1/4, and 1/16, the average latency of MEE-ElasticEdge presents a slight reduction. Bandwidth and latency have no apparent changes when the reserved region ratio is 1/16.

Experimental Summary: From the experimental results, we find that ElasticEdge has evident advantages of high-performance video transmission. Compared with the conventional live video framework, ElasticEdge has advantages as follows.

- 1) ElasticEdge can save bandwidth in multiple hardware environments, and transmission bandwidth can be reduced by up to approximately 60%. ElasticEdge has a noticeable edge over RTFace in terms of saving transmission bandwidth.
- 2) In different hardware environments, latency is reduced by approximately 10% with ElasticEdge compared to without ElasticEdge. In the case of more filtering processing, ElasticEdge has lower latency than RTFace.
- 3) In an environment of poor network conditions, the system with ElasticEdge can provide more reliable video stream transmission than the system without ElasticEdge.

- 4) Using ElasticEdge also reduces the cost of transmission bandwidth over 4G and 5G networks. Therefore, ElasticEdge can significantly reduce transmission bandwidth over different networks.
- 5) Under different hardware environments, using ElasticEdge can significantly reduce power consumption compared with not using ElasticEdge.

VII. RELATED WORK

Edge Computing: In the era of the IoE [48], [49], data generated by things are uploaded to the cloud center, which increases the network load and data processing latency [50]. Over the past decade, the cloud computing model has been widely used to process mass data in many areas [51], [52]. The paradigm of transferring data to the cloud center rarely impacts the user experience. In latency-sensitive applications, the limitation of the cloud computing model is evident [53]. The cloud model cannot satisfy real-time requirements [54], and the accumulation of multitasks significantly impacts the user experience. Edge computing allocates a portion of computing tasks to edge nodes close to the data source, thereby reducing the response time and network overload [55]. When processing latency-sensitive video analytics tasks, edge nodes can preprocess video data to release the burden on the network and cloud center [56], thereby improving the application response time.

Real-Time Video Analysis in Cloud and Edge Modes: There are several cloud- and edge-based real-time video

analysis systems in cloud computing [57]. Haigen *et al.* [58] presented a scheduling strategy for real-time video analysis. Anjum *et al.* [59] designed a cloud-based video analysis framework for scalability analysis in video streams.

In edge remote live video, Ananthanarayanan *et al.* [60] indicated that real-time video analysis is a killer application of edge computing. Bilal and Erbad [12] handled many challenges by edge computing-based interactive media applications. Sada *et al.* [61] introduced a distributed video analysis framework. Sun *et al.* [62] proposed mVideo, a video streaming platform based on mobile edge computing, capable of fully exploiting resources on collaborative edge and cloud nodes for video analysis. These authors also implemented a video usefulness (VU) model based on edge computing for a large-scale video surveillance system [63]. Edge computing is used to process video data close to sensors and medical equipment to achieve real-time performance [64], [65]. Liu *et al.* [66] combined data processing and edge computing on drones to save wireless bandwidth, demonstrating scalability and real-time performance. The previously mentioned work applied edge computing to perform video processing or intelligent processing at the edge node, achieving real-time processing. Some pieces of state-of-the-art video filtering techniques [67]–[69] are primarily used in specific environments where a clear and simple pattern of contextual irrelevance can be observed. These filtering methods are not dynamic. A context-aware periodical filtering mechanism is proposed to make decisions on video frames based on the application contexts [70]; however, this method cannot consider the ROI in the video stream of end users. It also uses a fixed value for the filter interval. These methods rarely consider the ROIs and the network conditions; then, it can aggregate the data transmission overhead under harsh network conditions. In this work, our proposed ElasticEdge performs regional filtering on video data according to the end-user requirements. ElasticEdge reduces the network transmission bandwidth and achieves remote live video stability by filtering out non-ROI regions.

Edge Computing and Remote Live Video Service: Edge computing exhibits advantages in remote video services. Taking telemedicine as an example, Xiao *et al.* [71] revealed that telemedicine uses edge computing and communication to offer high-quality medical services. In prior work, physiological characteristics were collected through sensors and medical equipment [72], [73]. However, large-scale medical video data cost high network bandwidth. Insufficient network bandwidth causes information transmission freezes, video blurring, and others, thereby causing a significant reduction in telemedicine. In remote emergency medical services, video analysis systems enable the doctors in the hospital to interact with edge devices equipped with video cameras at the accident site. Doctors control the remote camera position and obtain the video stream, assisting telediagnosis. Elgamal *et al.* [74] designed a telemedicine system, Teleconsultant, offering near real-time communication between paramedics and doctors. The paramedic wears a camera and communicates with the laptop in the ambulance via the ad hoc or Wi-Fi network. The laptop acts as a video stream server and delivers video to

doctors in the hospital. Doctors review the video from the accident site on a desktop installed with a stream client and in-stream processors. Schaer *et al.* [75] proposed a medical platform that leverages smart glasses as wearable cameras in emergency medical situations. With the platform, paramedics can access medical knowledge and obtain assistance from hospital specialists at the accident site via smart glasses-based video conferencing. Wu *et al.* [76] proposed an intelligent pre-hospital emergency medical services system, which explores wearable cameras, smart mobile devices, and video technology to improve the QoS of emergency medical services. Wearable sensing devices collect data, e.g., electrocardiogram, vital signs, and video clips of the patient, and data are transferred to the cloud. The hospital has prehospital visual data from the cloud and real-time video conferences for telemedicine.

Edge computing in telemedicine downregulates the amount of data transfer and alleviates bandwidth overload. The problem of data transmission quality is solved. In smart cities, edge computing with emerging monitoring software and hardware platforms has opened up tremendous possibilities. Khan *et al.* [77] introduced the role of edge computing to achieve the vision of a smart city, Chen *et al.* [78] proposed and deployed an edge computing-based distributed intelligent video surveillance system (i.e., DIVS), and Wang *et al.* [79] indicated that edge computing has great potential for future large-scale and distributed intelligent city monitoring applications. Most edge computing-based live video services are developed to implement a fixed processing mode at the edge node.

In summary, video transmission is a one-way mode in the remote emergency medical services system, and doctors mainly use interaction to view the video from sites. Still, they cannot define ROI in the video stream for elastic video processing, especially under harsh network conditions. ElasticEdge provides elastic video processing based on ROIs from remote doctors. In our ElasticEdge, video data can elastically facilitate processing, and the size of ROIs is elastic based on the user's requirements and under harsh network conditions, achieving high-quality live-video services.

Live Video Service Architecture: A conventional video service framework includes front-end cameras, transmission cables, and a back-end video platform. This framework has some issues, e.g., difficulty in analysis, passive monitoring, and strong dependence on manpower. In particular, by drawing upon humans to determine special circumstances, there is a lack of active handling and early warning for anomalies in the video. In addition, based on the conventional video service architecture, the ability to analyze, judge, and process the collected video is poor. Many novel architectures emerge to improve the mentioned scenarios.

At present, several edge computing-based solutions have been proposed for video processing. Satyanarayanan *et al.* [80] proposed conducting a video analysis at the network edge, reducing the need for cloud ingress bandwidth. EVAPS (i.e., public safety edge video analysis) [81] distributed the computing workload in edge nodes and clouds in an optimized manner. EVPAS reduces unnecessary data transmission, thereby saving energy on edge devices (e.g., cameras). Luo *et al.* [82]

proposed an edge security solution based on edge computing, i.e., EdgeBox, to detect security threat events in a near real-time manner and notify relevant parties when possible. However, many defects remain in the aforementioned works that should be improved. For example, users' requirements for real-time control of video data did not been taken into account. As a result, essential information cannot be promptly obtained through video data processing at the edge node according to the user's needs. Accordingly, in this study, ElasticEdge is proposed for live video, in which the edge node combines the edge device and the end-user end to perform comprehensive video processing for the video data.

VIII. CONCLUSION

In this article, we proposed an intelligent edge framework named ElasticEdge for live video analytics. ElasticEdge adjusts the end-user's real-time control on video analysis and enables the interaction between the edge device and the end user to avoid the end user passively being accepted video streams. ElasticEdge automatically selects the video data of user-defined ROI to facilitate the video data processing to achieve adjustment in the video stream. ElasticEdge transmits the video stream to the end users based on the tradeoff between the amount of video data and users' ROI to meet various network conditions. We implemented the prototype ElasticEdge system, on which we conducted extensive experiments to validate our proposed framework compared with RTFace. The results show that ElasticEdge has advantages in performance and energy consumption. This framework has significant dynamic performance, which can effectively adapt to various network conditions and reduce latency in harsh environments.

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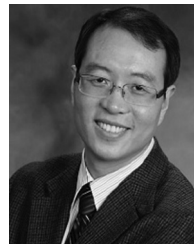
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